

SIT225 Data Capture Technologies

Assessment Pass Task 3P – Peer Review

Peer Review of Assessment 2 – Gordon

Reviewer: Deven Garg

Peer reviewed: Gordon

Date: 31 August 2026

The following feedback is based on the peer's Assessment 2 submission and addresses the eight review criteria specified in the SIT225 Task 3 peer review protocol.

Review Criterion	Peer Feedback
1. Use case clarity	The use case is clear and focuses on improving a low-cost anti-theft device by combining a PIR sensor with an LSM6DS3 accelerometer. The refined research question is more specific than the original problem because it focuses on reducing false alarms by combining different sensor information. The distinction between PIR detecting nearby human movement and the accelerometer detecting physical movement is easy to understand. However, the target of reducing false alarms by exactly 40% needs stronger justification because it is presented as a target without explaining how the value was selected. It would also help to define more clearly what counts as actual theft compared with normal activities such as typing, walking past the desk or accidentally touching the desk. Overall, the problem is researchable, but clearer operational definitions would make the study easier to reproduce.
2. Literature review rigour	The literature review includes two relevant anti-theft/security studies and identifies limitations in existing systems. It connects the literature to issues such as false alarms, sensor limitations and real-time monitoring. However, the literature base is relatively small, with only two main sources used to establish the research gap. Some claims about PIR false positives, accelerometer sensitivity and statistical requirements would benefit from direct citations to technical or peer-reviewed sources. The research gaps also move between vehicle security, smart homes and the proposed desk security device. Adding another source specifically related to multi-sensor fusion or PIR/accelerometer-based detection would strengthen the research justification.

3. Hypothesis quality	H1 is the strongest hypothesis because it provides measurable targets of greater than 95% theft detection and less than 5% false alarms when both sensors are used. The proposed 50 test runs also provide a practical way to compare the sensors. H2 is understandable because it compares short accidental bumps with longer movements associated with theft. However, there is an issue between H2 and the proposed 1 Hz sampling rate. If an accidental bump lasts for less than one second, a reading taken once per second could completely miss it. Gordon should either increase the sampling rate for this test or change H2 so that the events being investigated can reliably be measured at 1 Hz.
4. Sensor selection	The PIR sensor and LSM6DS3 accelerometer are appropriate choices because they measure different characteristics of a possible theft event. PIR detects nearby human movement while the accelerometer detects physical movement or shaking. Combining them is therefore useful for reducing false alarms. The use of acceleration magnitude from the X, Y and Z axes is also reasonable. However, the protocol could explain more clearly where the sensors are physically positioned and how their detection areas overlap. The reason why 1 Hz is sufficient should also be explained more clearly, particularly for short movements.
5. Ethical considerations	The ethical section is one of the stronger parts of the design because it recognises that motion information collected in a home can reveal people's routines even without names or other direct identifiers. The privacy notice, no-camera/no-microphone approach and defined retention periods are positive measures. However, the privacy notice refers to “facility managers” and “weekly summaries”, which does not seem to match the home-based testing described in the project. This should be changed to reflect the actual stakeholders. The consent process could also explain more clearly how family members can agree to participate and withdraw from testing.

6. Feasibility	The proposed experiment is generally feasible for a student because it uses relatively simple hardware, local CSV storage and a limited collection period. The 50 test runs are also manageable. However, the project includes several additional components such as Arduino Cloud monitoring, email/browser notifications, missing-data handling and statistical analysis. I recommend prioritising the core experiment first: comparing PIR alone, accelerometer alone and both sensors together. Cloud monitoring could be treated as an extension rather than being essential to validating the main hypotheses.
7. Strongest element	The strongest element is the decision to combine the PIR and accelerometer rather than relying on a single sensor. The two sensors observe different characteristics of an event, which gives the project a clear technical purpose. The proposed logic that an alert should require PIR movement and sustained item movement is a sensible approach to reducing false alarms. The project also has measurable performance targets rather than only demonstrating that the device works.
8. One improvement	The most important improvement would be to align the sampling rate, hypotheses and testing method. The current design uses 1 Hz sampling, but H2 claims that short bumps lasting less than one second can be distinguished from theft movements lasting several seconds. At one reading per second, a short bump could happen completely between two readings and never appear in the dataset. Increasing the sampling rate during controlled event testing, or redefining H2 around events that last long enough to be captured at 1 Hz, would make the hypothesis more scientifically defensible.

Acknowledgement

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